
PARIS AGREEMENT ON CLIMATE CHANGE: RISING SEA LEVELS THREATENING PHILIPPINE CITIES

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“Even if we stop all greenhouse emissions today – which is not exactly realistic – sea levels will continue to rise. That’s not to say that what we do doesn’t matter. In fact, it matters a great deal.”¹

Climate change is a significant issue in the Philippines that is already causing significant economic, social, and environmental impacts. The country is particularly vulnerable to the impacts of climate change due to its location and its high population density and one of the environmental issues is the sea level rise because of climate change.²

The Philippines aims to combat climate change and adapt to its effects. Thus, Philippines was a signatory to the Paris Agreement, a landmark treaty in the multilateral climate change process which is a binding agreement for all nations together to combat climate change and adapt to its effects.³

Moreover, the 1987 Philippine Constitution upholds preservation and protection of our patrimony in its Preamble, and the right to a balanced and healthful ecology enshrined in Article II, Section 16 which provides that *“The State shall protect and advance the right of the people to a balanced and healthful ecology in accord with the rhythm and harmony of nature.”⁴*

Moreover, the Supreme Court in the case of *Oposa v. Factoran* stated that *“the right to a balanced and healthful ecology need not even be written in the Constitution for it is assumed, like other civil and political rights guaranteed in the Bill of Rights, to exist from the inception of mankind and it is an issue of transcendental importance with intergenerational implications.”⁵*

Thus, the government must take steps to mitigate greenhouse gas emissions and adapt to the changing climate, but much more needs to be done to reduce the

¹ Sebesvari, Rita, Sea level rise is inevitable – but what we do today can still prevent catastrophe for coastal regions. (September 26, 2019) <https://theconversation.com/sea-level-rise-is-inevitable-but-what-we-do-today-can-still-prevent-catastrophe-for-coastal-regions-124129>

² Philippine Atmospheric Geophysical and Astronomical Services Administration (PAG ASA) (n.d.), <https://www.pagasa.dost.gov.ph/information/climate-change-in-the-philippines>

³ United Nations Framework Convention on Climate Change (UNFCCC) The Paris Agreement. (n.d.), <https://unfccc.int/process-and-meetings/the-paris-agreement>

⁴ CONST. (1987), art. II, §16 (Phil.).

⁵ *Oposa v. Factoran*, G.R. No. 101083, (July 30, 1993) (Phil.).

impacts of climate change on the country and its people. The Philippines must continue to prioritize climate action to build a more sustainable and resilient future.

Climate Change and its Impact in the Philippines

Today, the worst impacts are mostly felt in places where sea level rise has combined with sinking land, also known as subsidence. Subsidence is a natural process, but the weight of high-rise buildings and the emptying of underground water reservoirs tends to speed it up. That's why sinking coastal megacities may suffer a great deal from just 20-40cm of sea level rise.⁶

Climate change is having serious impacts in the Philippines such as increased frequency and severity of natural disasters, sea level rise, extreme rainfall, resource shortages, and environmental degradation.⁷ All of these impacts together have greatly affected the Philippines' agriculture, water, infrastructure, human health, and coastal ecosystems and they are projected to continue having devastating damages to the economy and society of the Philippines.⁸

According to the UN Office for the Coordination of Humanitarian Affairs (OCHA), the Philippines is one of the most disaster-prone countries in the world.⁹ The archipelago is situated along the Pacific Ocean's typhoon belt, leaving the country vulnerable to around 20 typhoons each year, a quarter of which are destructive.¹⁰ The December 2021 typhoon known colloquially as Typhoon Odette caused around a billion dollars (₱51.8 billion) in infrastructure and agricultural damages and displaced about 630,000 people. The United Nations estimated that Typhoon Odette affected the livelihoods of 13 million people, destroying their homes and leaving them without adequate food or water supplies.¹¹ More tragically, the

⁶ Sebesvari, Rita Sea level rise is inevitable – but what we do today can still prevent catastrophe for coastal regions (September 26, 2019), <https://theconversation.com/sea-level-rise-is-inevitable-but-what-we-do-today-can-still-prevent-catastrophe-for-coastal-regions-124129>

⁷ United States Agency for International Development (USAID) CLIMATE CHANGE RISK PROFILE PHILIPPINES (n.d.), https://www.climatelinks.org/sites/default/files/asset/document/2017_Climate%20Change%20Risk%20Profile_Philippines.pdf

⁸ *Id.*

⁹ UNOCHA "About OCHA in The Philippines" (March 2019) <https://www.unocha.org/philippines>

¹⁰ Asian Disaster Reduction Center Information on Disaster Risk Reduction of the Member Countries (n.d.) <https://www.adrc.asia/nationinformation.php?NationCode=608&Lang=en>

¹¹ ABS-CBN News "PAGASA redefines 'super typhoon', revises wind signals". (March 23, 2022) <https://news.abs-cbn.com/news/03/23/22/pagasa-redefines-super-typhoon-tweaks-wind-signals>

physical and economic repercussions of Typhoon Odette led to the death of over 400 people as of December 2021.¹²

Compounding these issues, the impacts of climate change, such as accelerated sea level rise, exacerbate the state's high susceptibility to natural disasters, like flooding and landslides.¹³ Aside from geography, climate change impacts regions with a history of colonization more intensely than regions without a history of colonization.¹⁴ Colonized regions experience the repercussions of climate change most jarringly "because of their high dependence on natural resources, their geographical and climatic conditions and their limited capacity to effectively adapt to a changing climate."¹⁵ Since low-income countries have a history of colonialism and resource exploitation, their environment lacks the diversity necessary to prevail against natural disasters.¹⁶ A lack of biodiversity reduces the resilience of a specific region, leaving them more susceptible to natural disasters and the effects of climate change. With its history of Spanish colonization, the Philippines is not environmentally nor economically equipped to overcome issues it is currently dealing with, such as natural disasters and climate change. This inability to recover exacerbates the problem, creating a cycle of environmental and economic devastation in the country.¹⁷

The Philippines said it was revising up its target to cut greenhouse gas emissions to a 75% reduction by 2030 under its commitment to the Paris Agreement on Climate Change, up from a target of 70% set four years ago.¹⁸

Finance Secretary Carlos Dominguez III, who also chairs the country's Climate Change Commission, described the new target as "ambitious", though some environmental activists called on the government to do more.¹⁹

¹² Typhoon Rai. Wikipedia https://en.wikipedia.org/w/index.php?title=Typhoon_Rai&oldid=1087095376

¹³ Perez, Rosa T.; et al., "Climate Change Impacts and Responses in the Philippines Coastal Sector" (August 1999), *Climate Research*, 12 (2/3): 97–107, Bibcode:1999ClRes..12...97P, doi:10.3354/cr012097, JSTOR 24866004

¹⁴ Smith, Paul. "Climate Change, Weak States and the "War on Terrorism" in South and Southeast Asia" (2007). *Contemporary Southeast Asia*. 29 (2): 264–285. doi:10.1355/CS29-2C. ISSN 0129-797X. JSTOR 25798831.

¹⁵ *Id.*

¹⁶ Das Gupta, Monica. "Population, Poverty, and Climate Change". (2014). *The World Bank Research Observer*. 29 (1): 83–108. doi:10.1093/wbro/lkt009. hdl:10986/22565. ISSN 0257-3032. JSTOR 24582389.

¹⁷ *Id.*

¹⁸ Reuters Philippines raises carbon emission reduction target to 75% by 2030 | Philippines raises carbon emission reduction target to 75% by 2030 | Reuters (2021) <https://www.reuters.com/business/environment/philippines-raises-carbon-emission-reduction-target-75-by-2030-2021-04-16/>

¹⁹ *Id.*

Breaking down the target, 72.29% is conditional on the support of climate finance, technologies and capacity development provided by developed countries, as prescribed by the Paris Agreement.²⁰

The Paris Agreement, which came into force in 2016, aims to transform the world's fossil-fuel-driven economy within decades and slow the pace of a global temperature increase to well below 2 degrees Celsius.

The Philippines ratified the agreement in 2017, allowing Manila access to the Green Climate Fund, which aims to channel billions of dollars to help poor nations tackle global warming.²¹

The remaining 2.71% of the target should be implemented mainly through domestic resources, the Department of Finance said in a statement.²²

Under a "business-as-usual" scenario, the government estimates the Southeast Asian country's overall emissions during the 2020-2030 period will be 3,340.3 metric tons of carbon dioxide equivalent (MTCO2e).²³

Greenpeace, however, called on the government to do more and for the Department of Energy to speed up the country's energy transition.

"Following the administration's pronouncements to fast-track renewable energy and for a moratorium on new coal proposals, the energy department should be doing much more to put their words into action," the environmental organization said in a statement.²⁴

Sea Level Rise

Globally, an estimated 150 million people live in communities that will be submerged in water by 2050. One of the problems of environmental issues is the sea level rise. The rise in sea level is an increase in the level of the world's oceans because of global warming. Burning fossil fuels is one of the causes of global warming because it releases carbon dioxide and other heat-trapping gases into the atmosphere. The oceans then absorb most of this heat. Sea levels are rising because of climate change. This rise is likely to accelerate over the coming century and continue for centuries. The impacts of sea level rise include permanent flooding (inundation) of low-lying

²⁰ Reuters Philippines raises carbon emission reduction target to 75% by 2030 | Philippines raises carbon emission reduction target to 75% by 2030 | Reuters (2021) <https://www.reuters.com/business/environment/philippines-raises-carbon-emission-reduction-target-75-by-2030-2021-04-16/>

²¹ *Id.*

²² *Id.*

²³ *Id.*

²⁴ *Id.*

areas, and increased frequency, extent and depth of tidal inundation. Sea level rise will also cause most sandy beaches to recede (where beaches will move further inland) and erode.²⁵

But the sea level rise we see today is the consequence of global warming that started from emissions released decades ago. Large bodies of water like the oceans have a great heat capacity – they warm up slowly and keep that temperature for a long time. This means that changes in sea level lag warming of the atmosphere.²⁶ It was forecasted that over 410 million people could be at risk from rising sea levels by 2100. The climate crisis has already caused global sea levels to rise by over 10cm between 1993 and 2024. In its Global Risks Report 2024, the World Economic Forum identified sea level rise from collapsing ice sheets as a key threat over the coming decade. Homes, livelihoods and ultimately lives are under threat from rising sea levels. By 2100, up to 410 million people could be at risk from coastal flooding as the climate crisis causes sea levels to rise even higher.²⁷

Impacts of Sea Level Rise in the Philippines

The Philippines is among the countries most affected by rising sea levels²⁸ and is not spared as it is also experiencing sea-level rise, which is threatening more than 40% of the population that live in coastal areas. The rising sea levels are causing increased flooding, erosion, and salinization of freshwater resources. These impacts have significant economic, social, and environmental consequences, including the displacement of people and destruction of coastal infrastructure. To address the impacts of climate change, the Philippine government has taken steps to mitigate greenhouse gas emissions and adapt to the changing climate. The country has committed to reducing its greenhouse gas emissions by 75% by 2030, which will require significant changes in the energy and transportation sectors.²⁹

The children are the most vulnerable. As of 2023, records from three decades show sea levels rising by 7 to 10 centimeters per decade along Philippine coasts, which is three times faster than the world average.³⁰

²⁵ World Economic Forum, Sea level rise: Everything you need to know (July 11, 2024)., <https://www.weforum.org/agenda/2024/07/rising-sea-levels-global-threat/>

²⁶ *Id.*

²⁷ *Id.*

²⁸ Garcia, Ma Angelica, "LOOK: How rising sea levels could affect PHL cities, towns by 2050" (November 4, 2019), <https://www.gmanetwork.com/news/scitech/science/714173/how-rising-sea-levels-could-affect-phl-cities-towns-by-2050/story/>

²⁹ *Id.*

³⁰ Ines, Jezreel "In troubled waters: Rising sea levels threaten sinking town's survival". (July 4, 2023), <https://www.rappler.com/environment/in-troubled-waters-rising-sea-levels-threaten-sinking-town-survival/>

Philippine coastlines, one of the longest in the world, stretch more than 36,000 kilometers. Coastal areas and their communities are threatened by rising sea levels and gradual submersion. These endanger the livelihood of fishing communities as well as the country's food security.³¹

The report of the United Nations-backed Intergovernmental Panel on Climate Change released in September showed that the world's oceans and cryosphere—glaciers and polar regions—are getting warmer, more acidic and less productive.

In only three decades, areas in the Philippines now home to around 6.8 million people will likely be inundated as the sea levels continue to rise.³² By the end of the century, land currently occupied by 8.6 million Filipinos could be lower than the height of average annual coastal flood. The figures are among the findings of the study produced by New Jersey-based science organization Climate Central.

The research, published in the journal *Nature Communications*, found that 250 million people across the globe currently live on land below current annual flood levels and 110 million live below the high tide line at present.³³

Using a more accurate way of calculating land elevation called CoastalDEM, the study of authors Scott Kulp and Benjamin Strauss suggests there are 5.4 million Filipinos already occupying land below annual flood levels. Annual flood level is used to denote the water level at the shoreline that local coastal floods exceed on average once per year.³⁴

Conservative Estimates. The new projections for the Philippines—an archipelago in the Pacific Ocean—are based on a relatively conservative scenario of expected changes. This assumes humans will moderately reduce warming emissions roughly in line with the 2015 Paris climate agreement's two-degree Celsius target and there will be a fairly stable Antarctica. The numbers also used by the authors were based on a 2010 population data—100 million.³⁵

³¹ Ines, Jezreel "In troubled waters: Rising sea levels threaten sinking town's survival". (July 4, 2023), <https://www.rappler.com/environment/in-troubled-waters-rising-sea-levels-threaten-sinking-town-survival/>

³² Philstar, In 30 years, rising seas will threaten Philippine cities, towns home to 6.8 million (November 6, 2019) <https://www.philstar.com/headlines/2019/11/06/1966508/30-years-rising-seas-will-threaten-philippine-cities-towns-home-68m#:~:text=Sea%20levels%20are%20projected%20to%20rise%20nearly%20half%20a%20meter>

³³ Kulp, Scott A, et al., New elevation data triple estimates of global vulnerability to sea-level rise and coastal flooding (October 29, 2019), <https://www.nature.com/articles/s41467-019-12808-z>

³⁴ *Id.*

³⁵ *Id.*

But assuming the high greenhouse gas-emissions continue unabated along with Antarctic instability, seven million Filipinos live on land that could be threatened by inundation by mid-century. By 2100, the number will rise to 13 million.³⁶

Seven million Filipino below high tide line by 2050. The new data showed that 4.3 million people in the Philippines live in places that are below the high tide line. Tideline is used to indicate the long-term average of the highest daily local tide level. *“This suggests that this land is protected today, possibly by coastal defenses. It is also possible that our data underestimate elevation in some places,”* Peter Girard, director of communications at Climate Central.³⁷

Even under a highly optimistic scenario, 5.2 million Filipinos will occupy land that will be underwater at high tide by 2050. The number will increase to 6.9 million by the end of the century.³⁸

If emissions continue to rise, land currently home to 5.4 million Filipinos will fall below the level of high tide by mid-century. By 2100, the number will double to 11 million.³⁹

Sea Level Rise affecting Iloilo City

The sea level in the Philippines is rising three times faster than the global average, putting many of its coastal villages in peril, according to a climate scientist of the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA).⁴⁰

*“Based on a report done by Pagasa, we saw that the sea-level rise of the Philippines is rising three times faster compared to the global average.” “So, this can cause inundation of low-lying areas, especially many of our countrymen who live on the beach.”*⁴¹

The World Bank said rising sea levels “create not only stress on the physical coastline, but also on coastal ecosystems.”⁴²

³⁶ Kulp, Scott A, et al., New elevation data triple estimates of global vulnerability to sea-level rise and coastal flooding (October 29, 2019), <https://www.nature.com/articles/s41467-019-12808-z>

³⁷ *Id.*

³⁸ *Id.*

³⁹ *Id.*

⁴⁰ Corales, Nestor PH Sea Level Rising 3 times faster than World Average (September 23, 2022), <https://newsinfo.inquirer.net/1669093/ph-sea-level-rising-3-times-faster-than-world-average-says-expert>

⁴¹ *Id.*

⁴² *Id.*

The urbanized center of Iloilo City suffers from the same diminishing rates, particularly in the easter coastal districts of La Paz to Villa Arevalo. Here, the highest rates of subsidence were found within Mandurriao and Molo Districts and La Paz District. Such subsidence has also caused worse flooding and tidal disruption in the city proper.⁴³ Iloilo City is not spared as sea levels are projected to rise nearly half a meter by 2100 if Earth warms 2 degrees Celsius above pre-industrial levels and 84 centimeter in a 3C to 4C world.⁴⁴

Sea Level Measurement

Sea level is the measurement of the sea's surface height. Between the 1800s and early 1990s, tide gauges attached to structures such as piers measured global sea level, as research organization the Smithsonian Institution explains. Now satellites carry out this task by bouncing radar signals off the ocean's surface.⁴⁵

Because local weather conditions and other factors can affect sea level, measurements are taken globally and then averaged out.

In 2021, scientists discovered a sea-level "fingerprint" from the Greenland ice sheet, confirming fears about the extent to which ice is melting. These fingerprints are "detectable patterns of sea level variability around the world resulting from changes in water storage on Earth's continents and in the mass of ice sheets", according to NASA.⁴⁶

The Greenland ice sheet is now losing around 30 million tons of ice an hour, reports the Guardian, and a recent study predicts its total collapse could happen by 2025.⁴⁷

Extent of the Sea Level Rise

With the ice sheet at "*a tipping point of irreversible melting*", scientists currently expect an unavoidable sea level rise of 1-2 meters.⁴⁸

⁴³Esquire Mag Philippine Cities Are Sinking Faster Than We Realize. What Now? (August 27, 2024), <https://www.esquiremag.ph/politics/news/land-subsidence-climate-change-study-philippine-cities-a1057-20240827?s=5s8nj6gf8k6qrh22s92ef389ln>

⁴⁴ *Id.*

⁴⁵ World Economic Forum Sea level rise: Everything you need to know. (July 11, 2024), <https://www.weforum.org/agenda/2024/07/rising-sea-levels-global-threat/>

⁴⁶ *Id.*

⁴⁷ *Id.*

⁴⁸ *Id.*

Global sea levels have already risen by over 10cm between 1993 and 2024, according to NASA, which says sea levels have been rising at unprecedented rates over the past 2,500 years.⁴⁹

The US Space Agency and other US government agencies warned in 2022 that levels along the country's coastlines could rise by another 25-30cm by 2050.⁵⁰

The global sea level has risen by about 21cm since records began in 1880. While measuring in centimeters or even millimeters might seem small, these rises can have big consequences. This is particularly true where storm surges sweep further inland than they would have previously.⁵¹

Causes of Sea Level Rise

Two main factors cause sea level rise. These are melting ice from glaciers, and seawater expanding because of rises in global temperatures, explains the US National Oceanic and Atmospheric Administration.⁵²

Greenland and Antarctica are losing around 270 billion and 150 billion tons of ice a year, respectively.⁵³ There are also likely negative feedback loops that could speed up glacier ice melt. For example, the Thwaites Glacier in Antarctica is disintegrating more quickly than anticipated. It's nicknamed the 'doomsday glacier' because sea levels could rise more than three meters without it and its supporting ice shelves.⁵⁴

Heat stored in the ocean is responsible for between a third and half of global sea level rise, NASA says. The past decade has been the ocean's warmest since at least 1800, and ocean temperatures reached a new high in 2023.⁵⁵ Because local weather conditions and other factors can affect sea level, measurements are taken globally and then averaged out.⁵⁶

⁴⁹ World Economic Forum Sea level rise: Everything you need to know. (July 11, 2024), <https://www.weforum.org/agenda/2024/07/rising-sea-levels-global-threat/>

⁵⁰ *Id.*

⁵¹ *Id.*

⁵² *Id.*

⁵³ *Id.*

⁵⁴ *Id.*

⁵⁵ World Economic Forum Sea level rise: Everything you need to know. (July 11, 2024), <https://www.weforum.org/agenda/2024/07/rising-sea-levels-global-threat/>

⁵⁶ *Id.*

Philippines as a signatory to the Paris Agreement on Climate Change

The Paris Agreement is a legally binding international treaty on climate change. It was adopted by 196 Parties at the UN Climate Change Conference (COP21) in Paris, France, on 12 December 2015. It entered into force on November 4, 2016.⁵⁷

Its overarching goal is to hold “*the increase in the global average temperature to well below 2°C above pre-industrial levels*” and pursue efforts “to limit the temperature increase to 1.5°C above pre-industrial levels.”⁵⁸

However, in recent years, world leaders have stressed the need to limit global warming to 1.5°C by the end of this century. That’s because the UN’s Intergovernmental Panel on Climate Change indicates that crossing the 1.5°C threshold risks unleashing far more severe climate change impacts, including more frequent and severe droughts, heat waves and rainfall. To limit global warming to 1.5°C, greenhouse gas emissions must peak before 2025 at the latest and decline 43% by 2030.⁵⁹

The Philippines has already taken steps to pursue national actions to address climate change through its intended nationally determined contribution (INDC).⁶⁰

Despite financial limitations, Philippines has already demonstrated its commitment to meeting its pledge with the implementation of the national climate action plan and the integration of climate change in national and local development plans and budgets. It will continue to cascade climate change mitigation and adaptation actions to sub-national levels and to invest in climate – resilient local economies, consistent with post-2015 international frameworks such as the 2030 agenda on sustainable development, Sendai framework for disaster risk reduction, and most especially, the Paris agreement on climate change.⁶¹

To secure a sustainable future, the Philippines is pursuing value-adding strategies to ensure the provision of ecosystem services and green growth to address pollution and environmental degradation. To this end, a greenhouse gas inventory management and reporting system is being developed to create a transparent, accurate and comparable baseline of emissions.⁶²

⁵⁷ United Nations Framework Convention on Climate Change (UNFCCC) The Paris Agreement. (n.d.), <https://unfccc.int/process-and-meetings/the-paris-agreement>

⁵⁸ *Id.*

⁵⁹ World Economic Forum Sea level rise: Everything you need to know. (July 11, 2024), <https://www.weforum.org/agenda/2024/07/rising-sea-levels-global-threat/>

⁶⁰ Department of Environment and Natural Resources (DENR) (n.d.), <https://ncr.denr.gov.ph/news-events/ph-signs-the-paris-climate-change-agreement/>

⁶¹ *Id.*

⁶² *Id.*

It has also submitted recently its articles of accession to the Doha amendment to the Kyoto protocol.

As a highly vulnerable country, Philippines has committed early to contributing its fair share to the global action consistent with the climate vulnerable forum's Manila-Paris declaration which calls for a mitigation goal of below 1.5 degrees Celsius and full decarbonization and 100% renewable energy production by 2050.⁶³

Too much is at stake. Developed countries therefore need to do more in terms of dramatically raising their NDC ambition compatible with the 1.5-degrees Celsius threshold, as well as raising funding contribution to the climate action plans of vulnerable countries, which require adequate, predictable and sustainable financing. The Philippines also urges developed and developing countries alike to take urgent pre-2020 action.⁶⁴

Climate Change Commission (CCC)

Climate Change Commission⁶⁵ is the lead policy-making body of the government tasked to coordinate, monitor and evaluate government programs and ensure mainstreaming of climate change in national, local, and sectoral development plans towards a climate-resilient and climate-smart Philippines.⁶⁶

Vision

The Commission envisions a climate-resilient and climate-smart Philippines with highly adaptive communities.⁶⁷

Mission

To take the lead in the development and mainstreaming of evidence-based climate adaptation and mitigation policies through optimum coordination among key stakeholders towards achieving a climate-resilient and climate-smart Philippines with healthy, safe, prosperous, and self-reliant communities.

The Climate Change Commission (CCC) has vowed to exert its utmost efforts to finalize expeditiously the Philippines' commitment to the Paris agreement to reduce carbon emissions and adapt to the impacts of climate change.⁶⁸

⁶³ Department of Environment and Natural Resources (DENR) (n.d.), <https://ncr.denr.gov.ph/news-events/ph-signs-the-paris-climate-change-agreement/>

⁶⁴ *Id.*

⁶⁵ Climate Change Commission (n.d.), <https://climate.gov.ph/about-us>

⁶⁶ *Id.*

⁶⁷ *Id.*

⁶⁸ De Vera-Ruiz, Ellalyn, CCC vows to fast-track PH commitment to Paris climate agreement (January 18, 2021), <https://mb.com.ph/2021/01/18/ccc-vows-to-fast-track-ph-commitment-to-paris-climate-agreement/>

*"The country's first Nationally Determined Contribution (NDC) under the Paris Agreement shall convey the shared vision and collective strategies of our sectors for the immediate pursuit of low carbon development and a sustainable future for the Filipino nation," the CCC said.*⁶⁹

*"It shall also communicate to the developed world our readiness for transformative green investments and finance, consistent with the President's call for climate justice," it noted.*⁷⁰

The CCC, pursuant to its mandate under the law, has spearheaded the development and finalization of the NDC through a whole-of-government-and-society approach.⁷¹

*"We shall ensure that the NDC reflects information that is not only evidence-based, but also verifiable and accurate," it said.*⁷²

To ensure the technical integrity of the NDC, the Commission noted that the national government, with support from development partners, has devoted expertise, time, and resources to strengthen the capacities of the lead sectoral agencies to undertake the NDC analytics.⁷³

Moreover, the CCC said government agencies have required more time to develop and enhance the sectoral policies and measures, and to determine the country's economy-wide target for emission reduction and avoidance.⁷⁴

However, some civil society organizations had earlier expressed their discontent on the country's unambitious proposed commitment to reduce greenhouse gas (GHG) emissions to address climate change. They were citing the CCC's Dec. 23, 2020 presentation of the country's three-page draft NDC text.⁷⁵

In the latest version, the Philippines aims to reduce its GHG emissions by 30 percent by 2040 from its business-as-usual scenario (2020-2040), with at least two percent of these reductions to be done unconditionally. The groups highlighted that the reduction of 30 percent of GHG by 2040 is much lower than what the country had

⁶⁹ De Vera-Ruiz, Ellalyn, CCC vows to fast-track PH commitment to Paris climate agreement (January 18, 2021), <https://mb.com.ph/2021/01/18/ccc-vows-to-fast-track-ph-commitment-to-paris-climate-agreement/>

⁷⁰ *Id.*

⁷¹ *Id.*

⁷² *Id.*

⁷³ *Id.*

⁷⁴ *Id.*

⁷⁵ De Vera-Ruiz, Ellalyn, CCC vows to fast-track PH commitment to Paris climate agreement (January 18, 2021), <https://mb.com.ph/2021/01/18/ccc-vows-to-fast-track-ph-commitment-to-paris-climate-agreement/>

committed to reduce in its Intended Nationally Determined Contribution or INDC in 2015. The Philippines then had committed to reduce by 70 percent its carbon emissions by 2030 to achieve a below two degrees Celsius increase in global temperatures.⁷⁶

Climate Finance

Climate finance refers to local, national or transnational financing—drawn from public, private and alternative sources of financing—that seeks to support mitigation and adaptation actions that will address climate change. The Convention, the Kyoto Protocol and the Paris Agreement call for financial assistance from Parties with more financial resources to those that are less endowed and more vulnerable. This recognizes that the contribution of countries to climate change and their capacity to prevent it and cope with its consequences vary enormously. Climate finance is needed for mitigation, because large-scale investments are required to significantly reduce emissions. Climate finance is equally important for adaptation, as significant financial resources are needed to adapt to the adverse effects and reduce the impacts of a changing climate.⁷⁷

In accordance with the principle of “common but differentiated responsibility and respective capabilities” set out in the Convention, developed country Parties are to provide financial resources to assist developing country Parties in implementing the objectives of the UNFCCC. The Paris Agreement reaffirms the obligations of developed countries, while for the first time also encouraging voluntary contributions by other Parties. Developed country Parties should also continue to take the lead in mobilizing climate finance from a wide variety of sources, instruments and channels, noting the significant role of public funds, through a variety of actions, including supporting country-driven strategies, and taking into account the needs and priorities of developing country Parties. Such mobilization of climate finance should represent a progression beyond previous efforts.⁷⁸

It is important for all governments and stakeholders to understand and assess the financial needs of developing countries, as well as to understand how these financial resources can be mobilized. Provision of resources should also aim to achieve a balance between adaptation and mitigation.⁷⁹

Overall, efforts under the Paris Agreement are guided by its aim of making finance flows consistent with a pathway towards low greenhouse gas emissions and climate-resilient development. Assessing progress in provision and mobilization of support is also part of the global stock take under the Agreement. The Paris Agreement

⁷⁶ De Vera-Ruiz, Ellalyn, CCC vows to fast-track PH commitment to Paris climate agreement (January 18, 2021), <https://mb.com.ph/2021/01/18/ccc-vows-to-fast-track-ph-commitment-to-paris-climate-agreement/>

⁷⁷ United Nations Framework Convention on Climate Change (UNFCCC) (n.d.) <https://unfccc.int/topics/introduction-to-climate-finance>

⁷⁸ *Id.*

⁷⁹ *Id.*

also places emphasis on the transparency and enhanced predictability of financial support.⁸⁰

Climate Financial Mechanism & Sources of Fund

To facilitate the provision of climate finance, the Convention established a financial mechanism to provide financial resources to developing country Parties. The financial mechanism also serves the Kyoto Protocol and the Paris Agreement.⁸¹

The Convention states that the operation of the financial mechanism can be entrusted to one or more existing international entities. The Global Environment Facility (GEF) has served as an operating entity of the financial mechanism since the Convention's entry into force in 1994. At COP 16, in 2010, Parties established the Green Climate Fund (GCF) and in 2011 also designated it as an operating entity of the financial mechanism. The financial mechanism is accountable to the COP, which decides on its policies, program priorities and eligibility criteria for funding.⁸²

In addition to providing guidance to the GEF and the GCF, Parties have established two special funds—the Special Climate Change Fund (SCCF) and the Least Developed Countries Fund (LDCF), both managed by the GEF—and the Adaptation Fund (AF) established under the Kyoto Protocol in 2001.⁸³

At the Paris Climate Change Conference in 2015, the Parties agreed that the operating entities of the financial mechanism – GCF and GEF – as well as the SCCF and the LDCF shall serve the Paris Agreement. Regarding the Adaptation Fund serving the Paris Agreement negotiations are underway in the Ad hoc Working Group on the Paris Agreement (APA).⁸⁴

Nationally Determined Contribution (NDC), the NDC Implementation Plan (NDCIP) 2020-2030⁸⁵

To further realize the Philippines' set ambition in the first Nationally Determined Contribution (NDC), the NDC Implementation Plan (NDCIP) 2020-2030 outlines key actions, phasing, costs, and implementation arrangements for delivering

⁸⁰ United Nations Framework Convention on Climate Change (UNFCCC) (n.d.) <https://unfccc.int/topics/introduction-to-climate-finance>

⁸¹ *Id.*

⁸² *Id.*

⁸³ *Id.*

⁸⁴ *Id.*

⁸⁵ Implementation Plan for the Republic of the Philippines Nationally Determined Contribution (NDC) (2020-2030), [https://climate.gov.ph/public/ckfinder/userfiles/files/Knowledge/The%20Philippines' %20NDC%20Implementation%20Plan%20-%2020072024.pdf](https://climate.gov.ph/public/ckfinder/userfiles/files/Knowledge/The%20Philippines'%20NDC%20Implementation%20Plan%20-%2020072024.pdf)

the Philippines' NDC under the identified policies and measures in the transport, energy, waste, industry, and agriculture sectors.⁸⁶

In 2021, the Philippines submitted its Nationally Determined Contribution (NDC) to the United Nations Framework Convention on Climate Change (UNFCCC). Through the NDC, the Philippines commits to a projected greenhouse gas (GHG) emissions reduction and avoidance of 75%, for 2020 to 2030, of which 2.71% is unconditional and 72.29% is conditional. This NDC Implementation Plan (NDCIP) sets out a roadmap and actions for implementing the Philippines' NDC.⁸⁷

It outlines policies and measures (PAMs) to reduce or avoid emissions in five sectors: agriculture, waste, industry, transport, and energy. NDC PAMs include actions to be implemented in each sector, with implementation overseen by four sector departments: the Department of Agriculture (DA), Department of Energy (DOE), Department of Environment and Natural Resources (DENR), and Department of Transportation (DOTr). The PAMs were developed through a bottomup approach, building actions and needs at the sector level, to drive necessary action at the national level.⁸⁸

Regular updates to the plan will align with modifications to the NDC, ensuring continued relevance and efficacy.⁸⁹

Nationally Determined Contribution Policies and Measures

The initial phase of mitigation actions, encapsulated as NDC PAMs, are articulated in this plan. These initiatives set the Philippines on a pathway to low-carbon growth, delivering, once fully implemented, an aggregate reduction of approximately 990 million metric tons of carbon dioxide equivalent (mmtCO₂e) against the baseline. The delivery of the PAMs thus ensures that the unconditional NDC target of 2.71% is met and considerably exceeded, while also contributing to the partial fulfillment of the conditional target.⁹⁰

As additional support becomes available, the PAMs will be further enhanced through periodic reviews, enabling reductions toward achieving the NDC's conditional target. This plan will then be updated to reflect changes to and strengthen current and future PAMs.⁹¹

⁸⁶ Implementation Plan for the Republic of the Philippines Nationally Determined Contribution (NDC) (2020-2030), <https://climate.gov.ph/public/ckfinder/userfiles/files/Knowledge/The%20Philippines'%20NDC%20Implementation%20Plan%20-%20072024.pdf>

⁸⁷ *Id.*

⁸⁸ *Id.*

⁸⁹ *Id.*

⁹⁰ *Id.*

⁹¹ *Id.*

Future projections

Future projections for the current trajectory of climate change predict that global warming is likely to exceed 3 °C, potentially 4 °C, by 2060.⁹² Specifically in the Philippines, average temperatures are "virtually certain" to see an increase of 1.8 to 2.2 °C.⁹³ This temperature increase will stratify the local climate and cause the wet and dry seasons to be wetter and drier, respectively.⁹⁴ Most areas in the Philippines will see reduced rainfall from March to May, while Luzon and Visayas will see increased heavy rainfall.⁹⁵ There will also be an increase in: the number of days that exceed 35 °C; that have less than 2.5 mm of rainfall; and that have more than 300mm of rainfall.⁹⁶ Additionally, climate change will continue to increase the intensity of typhoons and tropical storms. Sea levels around the Philippines are projected to rise 0.48 to 0.65 meters by 2100, which exceeds the global average for rates of sea level rise. Combined with sea level rise, this stratification into more extreme seasons and climates increases the frequency and severity of storm surge, floods, landslides, and droughts. These exacerbate risks to agriculture, energy, water, infrastructure, human health, and coastal ecosystems.

Climate Change: The “greatest threat affecting our nations and people”

Former President Rodrigo Duterte called on industrialized nations to pay smaller countries like the Philippines reparations for the damage caused by climate change.⁹⁷

“We produce the least but we suffer the most. We are really the one paying for it the most. The pollution produced by other countries, we are the ones who catch it,” he said.⁹⁸

Addressing world leaders at the United Nations General Assembly in New York, President Marcos amplified his predecessor’s point, calling climate change the “greatest threat affecting our nations and peoples.”

⁹² World Bank. Getting a Grip on Climate Change in the Philippines (June 25, 2013). <https://www.worldbank.org/en/country/philippines/publication/getting-a-grip-on-climate-change-in-the-philippines>

⁹³ *Id.*

⁹⁴ United States Agency for International Development (USAID) CLIMATE CHANGE RISK PROFILE PHILIPPINES. (n.d.), https://www.climatelinks.org/sites/default/files/asset/document/2017_Climate%20Change%20Risk%20Profile_Philippines.pdf

⁹⁵ *Id.*

⁹⁶ *Id.*

⁹⁷ Galvez, Daphne, Rich countries must pay for climate change damage – Duterte. (April 19, 2022) <https://newsinfo.inquirer.net/1584618/rich-countries-must-pay-for-climate-change-damage-duterte#ixzz8mBA6HJCM>

⁹⁸ *Id.*

*“The Philippines, for example, is a net carbon sink. We absorb more carbon dioxide than we emit. And yet, we are the fourth most vulnerable country to climate change,” he said.*⁹⁹

*“There is no other problem so global in nature that it requires a united effort, one led by the United Nations. The effects of climate change are uneven and reflect a historical injustice,” Marcos said.*¹⁰⁰

In response to calls from the leaders of developing nations, including the Philippines, United Nations Secretary General Antonio Guterres said that *“it was time for “meaningful action” on the issue of compensation for damage wrought by the climate crisis.*”¹⁰¹

Recommendations

The government has been implementing measures to increase the resilience of the population to the impacts of climate change. These measures include the construction of flood control infrastructure, disaster risk reduction policies, and programs to promote sustainable agriculture but more should be done.

Implementation of the Paris Agreement requires economic and social transformation, based on the best available science. The Paris Agreement works on a five-year cycle of increasingly ambitious climate action -- or, ratcheting up -- carried out by countries. Since 2020, countries have been submitting their national climate action plans, known as nationally determined contributions (NDCs). Each successive NDC is meant to reflect an increasingly higher degree of ambition compared to the previous version.¹⁰²

Recognizing that accelerated action is required to limit global warming to 1.5°C, the COP27 cover decision requests Parties to revisit and strengthen the 2030 targets in their NDCs to align with the Paris Agreement temperature goal by the end of 2023, considering different national circumstances.¹⁰³

In their NDCs, countries communicate actions they will take to reduce their greenhouse gas emissions to reach the goals of the Paris Agreement. Countries also

⁹⁹ Galvez, Daphne, Rich countries must pay for climate change damage – Duterte. (April 19, 2022) <https://newsinfo.inquirer.net/1584618/rich-countries-must-pay-for-climate-change-damage-duterte#ixzz8mBA6HJCM>

¹⁰⁰ *Id.*

¹⁰¹ *Id.*

¹⁰² *Id.*

¹⁰³ *Id.*

communicate in their NDCs actions they will take to build resilience to adapt to the impacts of climate change.¹⁰⁴

Long-Term Strategies: To better frame the efforts towards the long-term goal, the Paris Agreement invites countries to formulate and submit long-term low greenhouse gas emission development strategies (LT-LEDS).¹⁰⁵

LT-LEDS provide the long-term horizon to the NDCs. Unlike NDCs, they are not mandatory. Nevertheless, they place the NDCs into the context of countries' long-term planning and development priorities, providing a vision and direction for future development.¹⁰⁶

Paris Agreement support. The Paris Agreement provides a framework for financial, technical and capacity building support to those countries that need it,¹⁰⁷ to wit:

Finance. The Paris Agreement reaffirms that developed countries should take the lead in providing financial assistance to countries that are less endowed and more vulnerable, while for the first time also encouraging voluntary contributions by other Parties. Climate finance is needed for mitigation, because large-scale investments are required to significantly reduce emissions. Climate finance is equally important for adaptation, as significant financial resources are needed to adapt to the adverse effects and reduce the impacts of a changing climate.¹⁰⁸

Technology. The Paris Agreement speaks of the vision of fully realizing technology development and transfer for both improving resilience to climate change and reducing GHG emissions. It establishes a technology framework to provide overarching guidance to the well-functioning Technology Mechanism. The mechanism is accelerating technology development and transfer through its policy and implementation arms.¹⁰⁹

Capacity-Building. Not all developing countries have sufficient capacities to deal with many of the challenges brought by climate change. As a result, the Paris Agreement places great emphasis on climate-related capacity-building for developing countries and requests all developed countries to enhance support for capacity-building actions in developing countries.¹¹⁰

¹⁰⁴ United Nations Framework Convention on Climate Change (UNFCCC) The Paris Agreement. (n.d.) <https://unfccc.int/process-and-meetings/the-paris-agreement>

¹⁰⁵ *Id.*

¹⁰⁶ *Id.*

¹⁰⁷ *Id.*

¹⁰⁸ *Id.*

¹⁰⁹ *Id.*

¹¹⁰ United Nations Framework Convention on Climate Change (UNFCCC) The Paris Agreement. (n.d.) <https://unfccc.int/process-and-meetings/the-paris-agreement>

¹¹⁰ *Id.*

Tracking Progress. With the Paris Agreement, countries established an enhanced transparency framework (ETF). Under ETF, starting in 2024, countries will report transparently on actions taken and progress in climate change mitigation, adaptation measures and support provided or received. It also provides for international procedures for the review of the submitted reports.¹¹¹ The information gathered through the ETF will feed into the Global stock take which will assess the collective progress towards the long-term climate goals.¹¹² This will lead to recommendations for countries to set more ambitious plans in the next round.¹¹³

Current Achievements. Although climate change action needs to be massively increased to achieve the goals of the Paris Agreement, the years since its entry into force have already sparked low-carbon solutions and new markets. More and more countries, regions, cities and companies are establishing carbon neutrality targets. Zero-carbon solutions are becoming competitive across economic sectors representing 25% of emissions. This trend is most noticeable in the power and transport sectors and has created many new business opportunities for early movers.¹¹⁴ By 2030, zero-carbon solutions could be competitive in sectors representing over 70% of global emissions.¹¹⁵

Lastly, no matter how seemingly small your contributions are, in the grand scheme of things, they make a difference that could create a significant effect towards achieving your goal. When it comes to environmental issues, small efforts create ripples, which could eventually form waves and finally shake the core of every individual to stir a tide of lifestyle change for the betterment of Mother Earth. Anchored on the concept of “what goes around comes around” and the law of cause and effect.

Quoting a Miss Earth who was asked “*Who or what do you think is the biggest enemy of Mother Earth and why?*” To which, she replied, “*I believe that the real problem in this world is not climate change. The real problem is us because of our ignorance and apathy. What we have to do is to start changing our ways, to start recalibrating our minds and redirecting our steps because together, as a global community, our micro-efforts will have a macro-effect to help save our home, our planet.*”¹¹⁶ In our individual capacities, let us be advocates of the use of reusable, sustainable energy to reduce emissions in the atmosphere and, ultimately, lessen the effects of global warming.

¹¹¹ United Nations Framework Convention on Climate Change (UNFCCC) The Paris Agreement. (n.d.) <https://unfccc.int/process-and-meetings/the-paris-agreement>

¹¹² *Id.*

¹¹³ *Id.*

¹¹⁴ *Id.*

¹¹⁵ *Id.*

¹¹⁶ Libero- Cruz, Grace Metro Style: Miss Earth 2017 Karen Ibasco: “Our Micro-Efforts Will Have a Macro-Effect” (November 9, 2017) <https://metro.style/people/celebrities/miss-earth-2017-karen-ibasco-our-micro-efforts-wil/7340>

To emphasize, *“Our micro-efforts will surely have a macro-effect to help save our home, our planet.”*

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